

PH2456 T8 - Perturbation Theory

1) Modified square well potential



$$V(x) = 0 \quad \{0 \leq x \leq L\} \rightarrow V(x) = 0 \quad \{0 \leq x \leq \frac{L}{2}\}$$
$$= \infty \quad \text{otherwise} \quad = \epsilon \quad \{\frac{L}{2} \leq x \leq L\}$$
$$= \infty \quad \text{otherwise}$$

The unperturbed energy states are $\psi_n^0(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$.

The perturbed energy values are given by the V' projection

$$E_n^1 = \langle \psi_n^0 | V'(x) | \psi_n^0 \rangle$$
$$= \frac{2}{L} \int_{\frac{L}{2}}^L dx \quad \epsilon \sin^2\left(\frac{n\pi x}{L}\right).$$

$$= \frac{\epsilon}{L} \int_{\frac{L}{2}}^L dx \left[1 - \cos\left(\frac{n\pi x}{L}\right)\right]$$

$$= \frac{L}{2} \cdot \frac{\epsilon}{L}$$

$$= \frac{\epsilon}{2}$$

$$E_n = E_n^0 + E_n^1$$

↑
unperturbed

↑
first order energy
correction

2) Special Relativity Corrections

$$E_K = mc^2 \sqrt{1 + \frac{p^2}{m^2 c^2}} - mc^2 \quad (\text{Rel. kinetic energy})$$

Taylor expand: $\sqrt{1+x} \approx 1 + \frac{x}{2} - \frac{x^2}{8} + \dots$

$$E_K = mc^2 \left[1 + \frac{p^2}{m^2 c^2} \frac{1}{2} - \left(\frac{p^2}{m^2 c^2} \right)^2 \frac{1}{8} + \dots \right] - \cancel{mc^2}$$

$$= \frac{p^2}{2m} - \frac{p^4}{8m^3 c^2} \quad (\text{first order relativistic correction})$$

In the QHO Hamiltonian $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m \omega^2 \hat{x}^2 - \frac{1}{8} \frac{\hat{p}^4}{m^3 c^2}$

$$E_n^1 = \langle n | \frac{\hat{p}^4}{m^3 c^2} | n \rangle$$

Note that $\hat{p} = i \sqrt{\frac{m \hbar \omega}{2}} (\hat{a}^\dagger - \hat{a})$

So $\langle n | \frac{\hat{p}^4}{m^3 c^2} | n \rangle = \frac{m^2 \hbar^2 \omega^2}{32} \cdot \frac{1}{m^3 c^2} \langle n | (\hat{a}^\dagger - \hat{a})^4 | n \rangle$

$$= \frac{\hbar^2 \omega^2}{32 m c^2} \langle n | (\hat{a}^\dagger - \hat{a})(\hat{a}^\dagger - \hat{a})(\hat{a}^\dagger - \hat{a})(\hat{a}^\dagger - \hat{a}) | n \rangle$$

$$= \dots \langle n | (\hat{a}^\dagger \hat{a}^\dagger - \hat{a}^\dagger \hat{a} - \hat{a} \hat{a}^\dagger + \hat{a} \hat{a}) (\hat{a}^\dagger \hat{a}^\dagger - \hat{a}^\dagger \hat{a} - \hat{a} \hat{a}^\dagger + \hat{a} \hat{a}) | n \rangle$$

$$= \dots \langle n | \cancel{\hat{a}^\dagger \hat{a}^\dagger \hat{a}^\dagger \hat{a}^\dagger} - \cancel{\hat{a}^\dagger \hat{a}^\dagger \hat{a} \hat{a}} - \cancel{\hat{a}^\dagger \hat{a} \hat{a}^\dagger \hat{a}^\dagger} + \hat{a}^\dagger \hat{a} \hat{a} \hat{a}^\dagger$$

$$\quad - \hat{a}^\dagger \hat{a} \hat{a}^\dagger \hat{a} + \hat{a}^\dagger \hat{a} \hat{a} \hat{a}^\dagger + \hat{a} \hat{a}^\dagger \hat{a}^\dagger \hat{a} - \cancel{\hat{a} \hat{a}^\dagger \hat{a}^\dagger \hat{a}}$$

$$\quad - \cancel{\hat{a} \hat{a}^\dagger \hat{a} \hat{a}^\dagger} + \hat{a} \hat{a}^\dagger \hat{a} \hat{a}^\dagger + \hat{a} \hat{a}^\dagger \hat{a}^\dagger \hat{a} - \cancel{\hat{a} \hat{a}^\dagger \hat{a} \hat{a}^\dagger}$$

$$\quad + \hat{a} \hat{a} \hat{a}^\dagger \hat{a}^\dagger - \cancel{\hat{a} \hat{a} \hat{a}^\dagger \hat{a}^\dagger} - \cancel{\hat{a} \hat{a} \hat{a}^\dagger \hat{a}^\dagger} + \hat{a} \hat{a} \hat{a}^\dagger \hat{a}^\dagger | n \rangle$$

We require states to be orthogonal $\langle n | n' \rangle = \delta_{nn'}$

$$= \dots \langle n | \hat{a}^\dagger \hat{a} \hat{a} \hat{a}^\dagger + \hat{a}^\dagger \hat{a} \hat{a} \hat{a}^\dagger + \hat{a} \hat{a}^\dagger \hat{a} \hat{a}^\dagger + \hat{a} \hat{a}^\dagger \hat{a} \hat{a}^\dagger$$

$$\quad + \hat{a} \hat{a} \hat{a}^\dagger \hat{a}^\dagger + \hat{a} \hat{a} \hat{a}^\dagger \hat{a}^\dagger | n \rangle$$

$$a^\dagger a |n\rangle = n |n\rangle$$

$$a a^\dagger |n\rangle = (1+n) |n\rangle$$

$$\begin{aligned} a^\dagger a^\dagger a a |n\rangle &= a^\dagger a^\dagger a \sqrt{n} |n-1\rangle \\ &= a^\dagger a^\dagger \sqrt{n} \sqrt{n-1} |n-2\rangle \\ &= \sqrt{n} \sqrt{n-1} \sqrt{n-1} a^\dagger |n-1\rangle \\ &= n(n-1) |n\rangle \end{aligned}$$

After going through each one, we should find that

$$E_n^1 = \frac{\hbar^2 \omega^2}{32 m c^2} (6n^2 + 6n + 3)$$